The hawk-dove “game”

The simplest demonstration of John Maynard Smith’s evolutionarily stable strategy (ESS) concerns a hypothetical response to aggression known as the hawk-dove “game.” In this, individuals can either be hawkish and fight until badly injured, or dovish and posture, but then retreat. Hawks will outmatch doves, but could be seriously harmed in a fight with another hawk. Doves routinely escape injury, but waste time in posturing. Which strategy would be better for passing on genes? Maynard Smith and his collaborators devised a mathematical model to provide the answer, and—in this instance—being more hawkish than dovish emerged as the ESS. It predicts a ratio of seven hawks for every five doves, which is equivalent to any one individual being hawkish seven-twelfths of the time, and dovish five-twelfths of the time.

See also: Evolution by natural selection • The selfish gene • Predator–prey equations • Ecological niches • Trophic cascades • Biodiversity and ecosystem function